

ASSIGNMENT-2

Q1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

ANS:-

Devices Using SMPS

1. **Desktop Computer (Home)** – Uses an SMPS to convert AC electricity into stable DC voltages required by the motherboard, CPU, RAM, and other components.
2. **LED TV (Home)** – Uses an SMPS to efficiently convert AC power into the lower DC voltages required by its electronic circuits.
3. **Projector (College)** – Uses an SMPS to convert AC electricity into regulated DC power needed by its internal electronic components.

Devices Using UPS

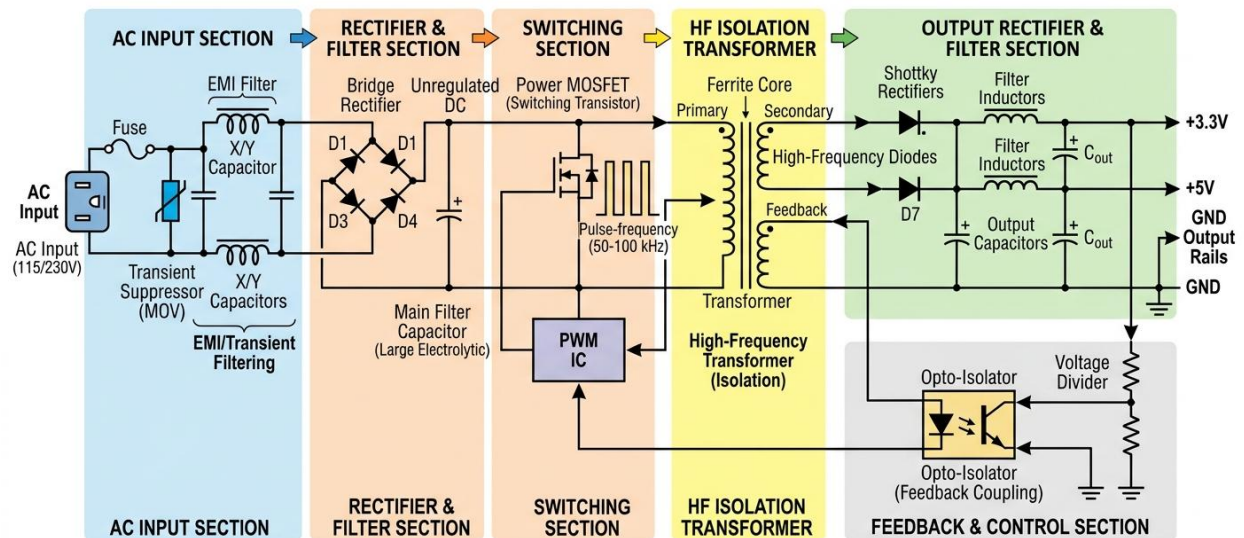
1. **Computer Lab PC (College)** – Uses a UPS to provide temporary battery backup during power cuts and protect the computer from sudden voltage changes.
2. **CCTV System (Home)** – Uses a UPS to keep the cameras and recording system working during a power failure.

Q2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io

ANS:-

INTERNAL SCHEMATIC OF A TYPICAL DESKTOP PC SMPS UNIT SIMPLIFIED DIAGRAM

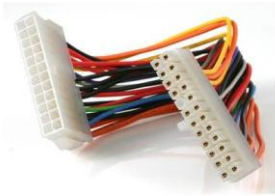


Q3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

ANS:-

Power Connectors

1. ATX 24-Pin Motherboard Connector



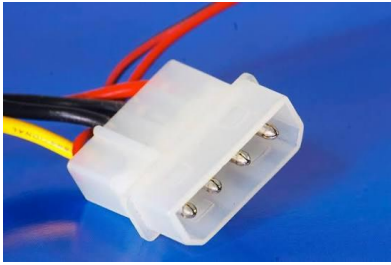
- **Devices powered:** Main motherboard power, including motherboard circuits and expansion-related power.
- **Safety tip:** Turn off and unplug the PC before connecting or disconnecting, and make sure the locking clip is correctly aligned.

2. SATA Power Connector



- **Devices powered:** SATA HDDs, SSDs, optical drives, and some other SATA-powered devices.
- **Safety tip:** Align the L-shaped connector correctly and never force it into the socket.

3. Molex Connector



- **Devices powered:** Older hard drives, optical drives, case fans, and other peripheral devices.
- **Safety tip:** Switch off and unplug the PC before connecting or removing the connector, and pull it out gently without twisting the wires.

4. PCIe 6/8-Pin Connector



- **Devices powered:** Dedicated graphics cards (GPUs) that require additional power.
- **Safety tip:** Use only the PCIe power cable intended for the graphics card and ensure the clip locks securely.

Q4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

ANS:-

Steps to Connect the SMPS in a New Gaming PC

1. **Turn off the power:** Switch off the SMPS and unplug the power cable from the wall socket before making any connections.
2. **Connect the motherboard:** Connect the **24-pin ATX cable** from the SMPS to the motherboard's main 24-pin power connector.
3. **Connect the CPU:** Connect the **4-pin or 8-pin EPS/CPU power cable** from the SMPS to the CPU power connector near the motherboard's processor socket.
4. **Connect the graphics card:** Connect the **PCIe 6-pin, 8-pin, or 6+2-pin PCIe cable** from the SMPS to the GPU's power connector.
5. **Connect SATA storage:** Connect a **SATA power cable** from the SMPS to the SATA HDD or SATA SSD.
6. **Connect other storage devices:** If an optical drive or other older peripheral is used, connect it with a **Molex power connector** when required.
7. **Check all connections:** Make sure every connector is fully inserted and its locking clip is securely fastened.
8. **Manage the cables:** Route and tie the cables neatly so they do not touch fans or block airflow.
9. **Final safety check:** Ensure no loose wires are present and verify that each cable is connected to the correct socket.
10. **Power on the PC:** Connect the SMPS power cable to the wall socket, switch on the SMPS, and then start the computer to check that all components receive power.